

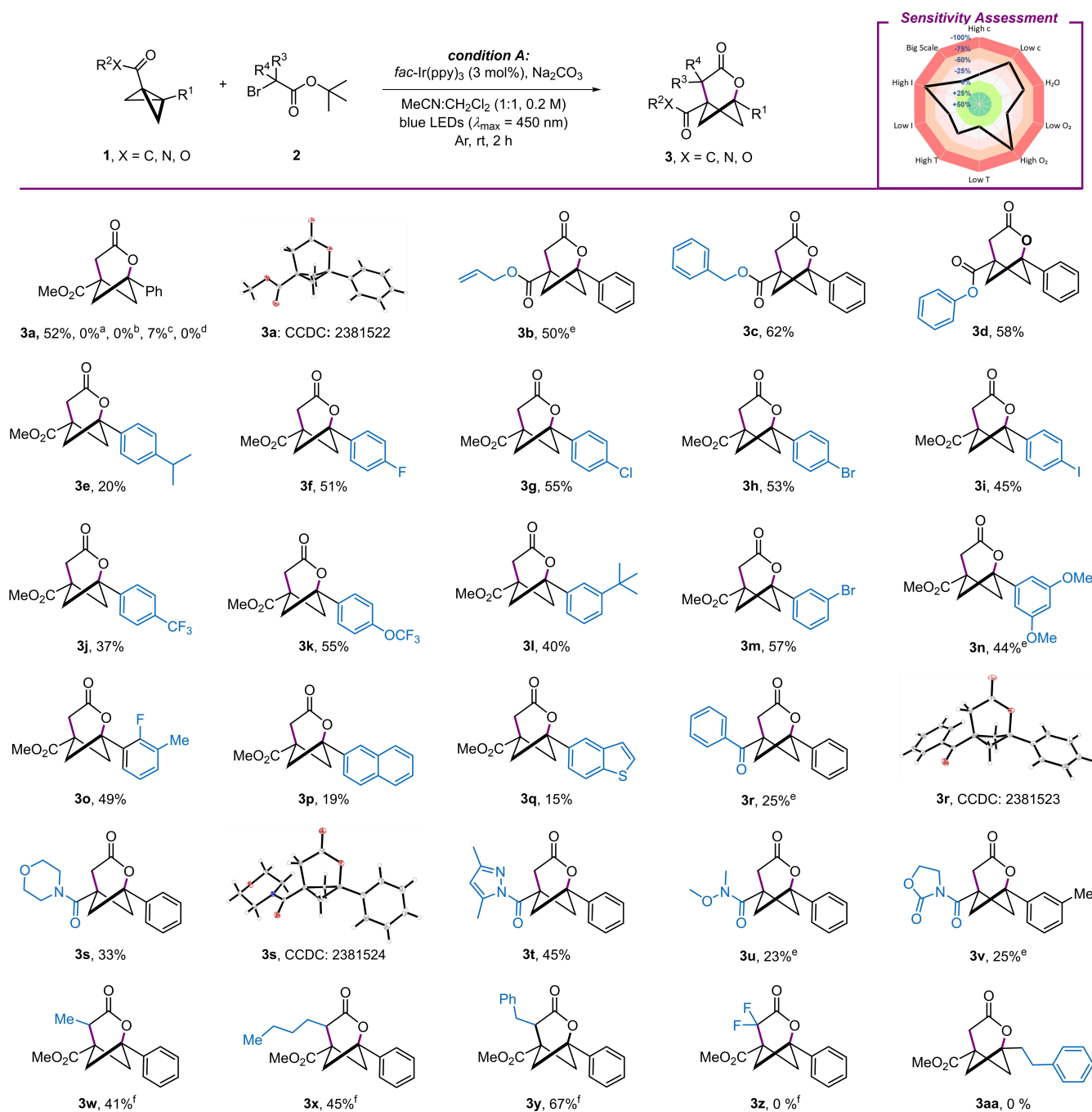


Figure 2. Substrate scope. **1** (0.2 mmol, 1.0 equiv.), **2** (1.1 equiv.), Na₂CO₃ (2.0 equiv.), *fac*-Ir(ppy)₃ (3 mol%), MeCN:CH₂Cl₂ (1:1, 0.2 M), at rt (30 °C to 35 °C) with blue LEDs (450 nm, 30 W) for 2 h under argon. Isolated yields are displayed. [a] Used *tert*-butyl chloroacetate instead of **2a**. [b] Used bromoacetic acid instead of **2a**. [c] Used *tert*-butyl iodoacetate instead of **2a**. [d] Used methyl bromoacetate instead of **2a**. [e] MeCN was the solvent. [f] 5 equiv. of **2** was used, with 16 h reaction time. The structures of the products were confirmed by X-ray crystallography analysis for **3a**, **3r**, **3s**. Reaction condition-based sensitivity assessment (radar graph).

decreased compared to standard conditions. Additionally, different types of ester groups yielded the desired cycloadducts with good yields (**4b–4d**). Even more interesting is that BCB containing a terminal alkene moiety was still well-tolerated in this reaction (**4b**). This suggests that the reaction can efficiently distinguish between different types of double bonds (according to the proposed mechanism in

Figure 5F). Next, except for the *para*-iodo-substituted giving 32 % yield (**4i**), BCB with phenyls bearing *para*- or *meta*-substituted electron-withdrawing or electron-donating groups, as well as disubstituted phenyls, all served as suitable substrates, providing the corresponding products in good yields (**4e–4h**, **4j–4o**). This indicates a broader range of functional group compatibility than 1,3-cyclization reac-